ELECTRODE ARRAY & PINOUT GUIDE · SKU PIA24027P

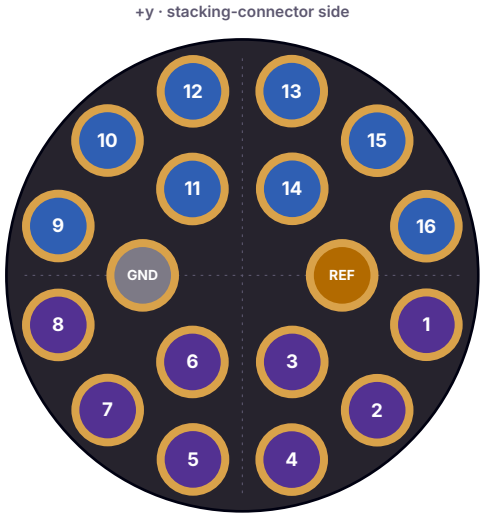
Pogo-Pin Map, Jumpers & Placement

Which pogo pin is which channel, where the reference and ground sit, how the jumpers are set, how the XIAO connects, and where to place the sensor on the head.



16	REF	GND	5.38	No clips
Signal pins	In the cluster	In the cluster	mm pitch, hexagonal	No other electrodes

01 THE POGO-PIN ARRAY



Pins	Count	Purpose
Signal 1–8	8	Converter #1
Signal 9–16	8	Converter #2
REF	1	Common reference (AINREF) for all 16 channels
GND	1	Ground

Reference and ground are inside the same cluster, so the whole montage sits on one spot of the scalp. Pin size Ø4 × 11 mm, spring-loaded.

● ch 1–8 ● ch 9–16 ● REF ● GND

Figure 1 — Pin-side view (looking at the pins), +y toward the stacking connector.

02 CHANNEL POSITIONS

Ch	x (mm)	y (mm)	Ring	Ch	x (mm)	y (mm)	Ring
1	+10.14	-2.72	outer	**9**	-10.14	+2.72	outer
2	+7.43	-7.43	outer	**10**	-7.43	+7.43	outer
3	+2.75	-4.77	inner	**11**	-2.75	+4.76	inner
4	+2.72	-10.15	outer	**12**	-2.72	+10.14	outer
5	-2.72	-10.14	outer	**13**	+2.72	+10.15	outer
6	-2.75	-4.77	inner	**14**	+2.75	+4.77	inner
7	-7.42	-7.42	outer	**15**	+7.43	+7.42	outer
8	-10.14	-2.73	outer	**16**	+10.14	+2.71	outer

Use these to draw a local topographic map. Channels 9–16 mirror channels 1–8 through the centre (x, y → −x, −y).

For research and education only. Not a medical device. Specifications subject to change.

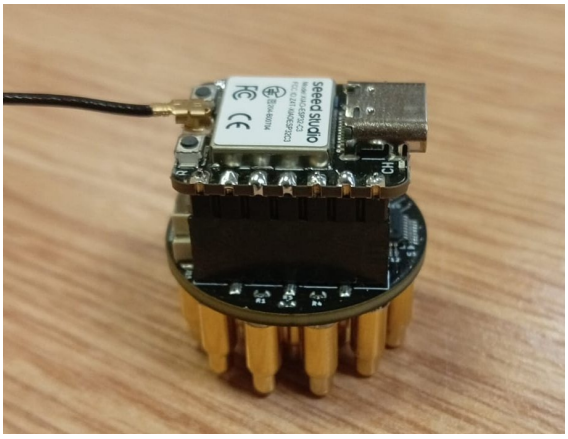
03 REFERENCE, GROUND AND STACKING JUMPERS

Jumper	Factory setting	What it does
AINREF_POGOPIN_ON	Fitted	Uses the REF pogo pin in the cluster (single-board use)
GND_ON	Fitted	Connects the GND pogo pin
AINREF_STACK_ON	Not fitted	Shares the reference through the stacking connector
XTAL_ON · SYNC_ON	Fitted	Clock and sample synchronisation (leave as supplied)
STACK_1	6-pin connector	Links stacked boards for a shared reference, ground and sync

CAUTION Stacking.
Fit AINREF_STACK_ON only when a stacking cable actually connects two boards. A loose cable on the reference line acts as an antenna and adds noise.

04 XIAO PIN MAP

XIAO pin	Signal
D0	Chip select, ADC 1 (ch 1–8)
D1	DRDY from ADC 1 (sync master)
D2	RESET, both ADCs
D3	Chip select, ADC 2 (ch 9–16)
D6	CLKOUT
D8 · D9 · D10	SPI SCLK · MISO · MOSI
5V · GND · 3V3	Power — must line up with the board labels

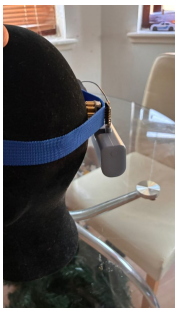


XIAO mounted the right way round, antenna clipped on.

WARNING Orientation.
A XIAO mounted 180° the wrong way still flashes and appears on Bluetooth, but sends no EEG (LED fast double-blink). Match the 5V / GND / 3V3 labels.

05 WHERE TO PLACE IT

Spot	10-20	Good for
Top-back of the head	Pz	Alpha rhythm (eyes closed) — default
Back of the head	Oz, O1, O2	Alpha, visual activity
Top of the head	Cz	Motor activity
Forehead	Fp1, Fp2	Blinks and eye movement (artifact demo)



Over the back of the head.



Held by a strap with the power bank.

Part the hair under the cluster, press the board **straight down**, hold it with a strap and sit still. Good contact on all 18 pins matters most.

06 RELATED DOCUMENTS

OC16-DS-001 Product Datasheet	**OC16-MD-002** Mechanical & Dimensional Drawing	**OC16-PCB-003** PCB & Manufacturing Specification
OC16-QSG-005 Quick Start & Safety Guide	**OC16-SW-006** Bluetooth Interface & Software Guide	